

Question Paper

Exam Date & Time: 30-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025

Ethical Hacking [NPT_4125]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Implement how the OSI and TCP/IP models differ in defining security boundaries for ethical hacking. (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Implement how TCP and UDP behaviors influence port-scanning strategies used by ethical hackers. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Implement the significance of IPv6 adoption in strengthening Internet-scale security. (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Implement how reconnaissance tools like Whois, Netcraft and DNSenum help attackers map a target system. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Explain the differences between NMAP and Nessus for vulnerability assessment and scanning accuracy. (5)
A)

Course Outcome: "CO2" / Cognitive Level: "Understand"
- B) Analyze how ARP spoofing enables man-in-the-middle attacks and suggest preventive countermeasures. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 4) Examine the role of digital certificates in SSL/TLS and their vulnerabilities to man-in-the-middle attacks. (5)
A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Analyze how DNS spoofing compromises domain resolution and user trust. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 5) Evaluate how the Metasploit framework automates exploitation and helps analyze system weaknesses. (5)
A)

- B) Analyze the effectiveness of countermeasures in case studies involving SQL-injection and XSS attacks. (5)

-----End-----

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4125:Ethical Hacking

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	15
3	3	CO 3	10
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4125:Ethical Hacking

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	4	20(40%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)